MILITARY SPECIFICATION

TEST SET, GUIDED MISSILE LAUNCHER
AN/ASM-20

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for flight line use and shall perform accurate and safe operational tests on launchers which launch the Sidewinder Missiles, Mark 2, Mark 29, and Mark 30.

1.2 Classification - The Test Set, Guided Missile Launcher AN/ASM-20 shall consist of the following items:

Items	Type Description	Appl. Para.
Test Set, Guided Missile Launcher	TS-1436/ASM-20	3.4.1
Cable, Special Purpose, Electrical	CX-6321/ASM-20	3.4.2
Adapter, Connector	U-213/U through U-216/U	3.4.3
Case, Test Set	CY-2943/ASM-20	3.4.4
Gauge, Pressure, Dial Indicating	MX-3298/ASM-20	3.4.5
Wrenches, HEX	58A5A294-12 & 60A5405	3.4.6

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682 Electron Tubes and Transistors, Choice and Application of

FSC 4940

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use With Electronic and Fire Control Systmes, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionics Equipment, General Specification for

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MS-17323	Mater, Time Totalizing

DRAWINGS

Bureau of Naval Weapons

58A5A294-12
60A54C5
60A54D6
50A54C81
60A54187
60A54C92
60A54C98
60A54D116

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2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Non repairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4632. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 75 pounds.

3.2.2 Reliability

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 150 hours without the necessity for readjustment of any controls which are inaccessible to the operator

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during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 200 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3

3.2.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with MIL-M-7793. Meters shall be included in the following units.

Unit	Type of Meter
TS-1436/ASM-20	MS17323-1

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the lower edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient

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3.2.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 2 equipment, except as modified herein.

3.2.10 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 10 minutes under standard conditions and 20 minutes at extreme service conditions.

3.2.11 Primary Input Power Requirements - The test equipment shall be supplied all necessary voltages for its operation from the missile launcher when attached to the aircraft power source.

3.2.11.1 Voltage Protection - The equipment shall not be damaged when exposed to transient voltages as defined in Figure 2 of MIL-STD-704, and shall automatically resume normal operation when the voltage returns within limits of normal operation.

3.2.12 Standby Provisions - The equipment shall provide for a standby position whereby rated voltages shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.3 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.4 Detail Requirements

3.4.1 Test Set, Guided Missile Launcher TS-1436/ASM-20 - The Test Set shall meet the following requirements:

3.4.1.1 Function - The Test Set shall be capable of determining that all necessary voltages and signals required by the Sidewinder Missile are available and of the proper level or value, and that all signal and missile data required by the launcher are similated and returned to the launcher.

3.4.1.2 Form Factor - The Test Set shall be enclosed in an instrument case and the maximum dimensions shall be 7 inches wide by 5 - ½ inches high by 8 - 3/4 inches long.

3.4.1.3 Weight - The weight of the test set shall not exceed 10 pounds.

3.4.1.4 Contents of Unit - The Test Set shall contain the necessary circuits, instrumentation and front panel components as follows:

(1) Microammeter (MI) - The meter shall register at the specified color area with respect to selector switch (S1) position, and the input voltage quantities and their tolerances.

(2) Selector Switch (S1) - The selector switch selects the launcher circuit to be checked and indicates the voltage or signal on meter MI as applicable for the (42) forty-two positions of S1.

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(3) Missile ID Switch (S2) - The Missile ID switch is a three position toggle switch used to select relays in the aircraft by missile type. The up position (IR) is used when testing MK29(IR) missile circuits, the middle position is used when testing 1 and 1A Missile circuits, and the down position is used when testing MK30(SR) missile circuits.

(4) Pilot Light - The pilot light should be lit during all positions of S1 except "off". If pilot light is not lit, the readings of M1 will probably be incorrect.

(5) Receptacle (J1) - Connects the Cable CX-6321/ASM-20 to the missile launcher under test.

3.4.1.5 Missile Launcher Circuit Tests - The Test Set, TS-1436/ASM-20, shall measure and register the individual voltage quantity with respect to pin 31 (ground) and the specified pins of the connector J101 of the Test Set. Cable Assembly, Special Purpose Electrical CX-6321/ASM-20 shall be the link between the Test Set and the launcher and aircraft circuitry when performing the missile launcher circuit tests. Connector P101 of Cable Assembly, CX-6321/ASM-20 shall be connected to J101 of the Test Set; P102 shall be connected to J102 of Adapter, Connector U-213/U or U-216/U; and the Lug Handle shall be connected to the forward (fuze) and aft (motor ignitor) contact points of the launcher. Adapter, U213/U or U-216/U shall be connected to the connector in the nose of the launcher.

(1) Adapter, connector U-213/U shall be used when testing launchers that launch the Sidewinder Missile Mark 2.

(2) Adapter, connector U-216/U shall be used when testing launchers that launch the Sidewinder Missile Mark 29 and Mark 30.

(3) Gauge, Pressure, MX-3298/ASM-20 shall test the nitrogen pressure system of the Mark 29 Launcher when inserted into the umbilical connector of the launcher.

(4) The "pins" referenced in the following switch positions are pins of connector J101 of the Test Set TS-1436/ASM-20.

3.4.1.5.1 Switch Position 1, Phase 1, 30 volts rms, 400 cps (S) - The Test Set shall measure the line to line voltage, 30 volts rms, 400 cps between pins 12 and 24. 25 volts rms, 400 cps, shall cause the meter to register at the left edge of the yellow area. 34.7 ± 1.4 volts rms, 400 cps, shall cause the meter to register at the right edge of the yellow area. The input resistance between pins 12 and 24 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.2 Switch Position 2, Phase 2, 30 volts rms, 400 cps (S) - the Test Set shall measure the line-to-line voltage, 30 volts rms, 400 cps between pins 12 and 25. 25 volts rms, 400 cycles shall cause the meter to register at the left edge of the yellow area. 34.7 ± 1.4 volts rms, 400 cps shall cause the meter to register at the right edge of the yellow area. The input resistance between pins 12 and 25 shall be

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75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.3 Switch Position 3, 30V, 400 cps Phase Sequence Check(S) - The Test Set will test for positive phase sequence using lines 12, 24, and 25. 25 VAC, rms ± 1.0 volt 400 cps, line-to-line shall when in positive sequence, cause the meter to read at the left edge of the yellow area. If the sequence is reversed the meter shall read below the yellow area. 34.7 ± 2.0 volts rms, 400 cps, line-to-line shall, when in positive sequence, cause the meter to read at the right edge of the yellow area.

3.4.1.5.4 Switch Position 4, Tuning Motor and AFC Meter (S) - The Test Set shall measure a voltage of 10.90 volts rms, 400 cps at pin 11 with pin 31 as ground reference. 8.75 volts rms, 400 cps shall cause the meter to register at the left edge of the yellow area. 11.93 volts ± 0.50 volts rms, 400 cps shall cause the meter to register at the right edge of the yellow area. The input impedance at pin 11, when position 4 is being tested, shall be 260 ohms at a temperature of 77° F + 20° F. The input at position 4 is cause by n output which is present on pin 23 of the Test Set when the selector swith is on position 4. The output signal shall be 0.87 volts peak-to-peak square wave, 400 cps, Phase I, superimposed on 4.90 volt dc nominal. The load appearing in the launcher will be 3000 ohms ± 10% to the ac signal and 5000 ohms ± 5% to the dc signal.

3.4.1.5.5 Switch Position 5, 6 KC, 134 volts rms Square Wave (S) - The Test Set shall measure the voltage at pin 28 of the Test Set with pin 31 as ground reference. 122 volts rms, ± 2 volts rms square wave, 6000 cps ± 500 cps shall cause the meter to read at the left edge of the yellow area. Nominal rise time, t_r, measured from 10% to 90% points is 15 usec. Nominal fall time, t_f, measured from 10% to 90% points is 20 usec. 147 ± 2 volts rms, square wave, 6000 ± 500 cps shall cause the meter to read at the right edge of the yellow area.

3.4.1.5.6 Switch Position 6, Missile B+ (AI) - The Test Set shall measure the voltage at pin 5 with pin 31 as ground reference. +155 volts dc (Positive) shall cause the meter to register at the left edge of the yellow area. 213 ± 7 volts dc shall cause the meter to register at the right edge of the yellow area. The input impedance at pin 5 shall be 1200 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.7 Switch Position 7, Automatic Gain Control, Disabling (A) - The Test Set shall measure a nominal voltage of 175 volts dc through a 10K resistance (located in the launcher) at pin 3 of the test set with pin 31 as ground reference. +148 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 204 volts ± 7 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 3 shall be 200K or greater at a temperature of 77° F ± 10° F.

3.4.1.5.8 Switch Position 8, Motor Drive B+ (A) - The unit shall measure the voltage at pin 6 with pin 31 as ground reference. +155 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 213 ± 7 volts dc shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 6 shall be 1200 ohms ± 4% at a temperature of 77° F ± 10° F.

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3.4.1.5.9 Switch Position 9, Motor Drive Filamentw (A) - The Test Set shall measure the voltage at pin 32, with pin 31 as ground reference. 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 32 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.10 Switch Position 10, Missile Filaments (AI) - The Test Set shall measure the voltage between pin 9 and pin 31. 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 9 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.11 Switch Position 11, Servo Heaters (AIS) - The Test Set shall measure the voltage at pin 16 with pin 31 as ground reference. 22.5 golts rms, 400 cps or 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input impedance or resistance at pin 16 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.12 Switch Position 12, Disabling B+ (AIS) - The Test Set shall measure the voltage at pin 7 with pin 31 as ground reference . +155 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 213 ± 7 volts dc shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 7 shall be 1200 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.13 Switch Position 13, 400 cps Audio Signal and Phase 1, 115 volt rms, 400 cps (AIS) - The Test Set shall measure the voltage at pin 30 with pin 31 as ground reference and check the operation of the pilot light. 101 ± 2.0 volts rms, 400 cps shall cause the meter to register at the left edge of the yellow area. 123 ± 2.5 volts rms, 400 cps shall cause the meter to register at the right edge of the yellow area. Voltage within this range shall cause the pilot light to glow. Input impedance shall be 5600 ohms minimum. Approximately 1 volt rms, 400 cps shall be present at pin 18. The output impedance at pin 18 shall be 150 ohms nominal.

3.4.1.5.14 Switch Position 14, Noncontinuity Between Firing Lines (non-short) (A) - The Test Set shall check the circuit continuity between the forward (fuze) and rear (propulsion) contacts of the launcher. Open circuit shall cause the meter to register in the yellow area. Short circuit shall cause the meter to register above the yellow area.

3.4.1.5.15 Switch Position 15, Uncage Relay Closed (AIS) - The Test Set shall check the circuit continuity between pins 2 and 10. Open circuit shall cause the meter to register in the black area. Short circuit shall cuase the meter to register in the yellow area. This indicates that the uncage relay is closed.

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3.4.1.5.16 Switch Position 16, Uncage Relay Open (AIS) - The Test Set shall check the circuit continuity between pins 2 and 10. Open circuit shall cause the meter to register in the red area. Short circuit shall cause the meter to register above the red area.

3.4.1.5.17 Switch Position 17, Gas Grain (AIS) - The Test Set shall measure the voltage at pin 26, with pin 31 as ground reference. 17.5 volts rms, 400 c;s or 17.5 voltd dc (positive) shall cause the meter to register at the left edge of the red area. 31 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the red area. The pnput impedance at pin 26 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.18 Switch Position 18, Launch Latch (IS) - The Test Set shall measure the voltage at pir 32, with pin 31 as ground reference. 17.5 volts rms, 400 cps or 17.5 volts dc (positive) shall cause the meter to read at the left edge of the red area. 31 ± 1. 2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the red area. The input impedance at pin 26 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.5.19 Switch Position 19, Fuze (A) - The Test Set shall measure the voltage at pin 19, with pin 31 as ground reference. 17.5 volts rms, 400 cps or 17.5 volts dc (positive) shall cause the meter to register at the left edge of the red area. 31± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the red area. The input resistance at pin 19 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F. The Test Set shall provide at pin 4 a voltage signal at a frequency of 6150 cps ± 200 cps.

3.4.1.5.20 Switch Position 20, Power Change Over and Fuze (IS) - The Test Set shall measure the voltage at pin 17 with pin 31 as a ground reference. 17.5 volts rms, 400 cps or 17.5 volts dc (positive) shall cause the meter to register at the left edge of the red area. 31 ± 1.2 volts rms, 400 cps or 3; ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the red area. The input impedance at pin 17 shall be 75 ohms ± 4% at a temperature of 77° F. The Test Set shall provide at pin 4 a voltage signal at a frequency of 6150 cps ± 200 cps.

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3.4.1.5.21 Switch Position 21, Propulsion (AIS) - The Test Set shall measure the voltage at pin 21 with pin 31 as ground reference. 17.5 volts rms, 400 cps or 17.5 volts dc (positive) shall cause the meter to register at the left edge of the red area. 31 ± 1.2 volts rms, 400 cps, or 31 ± 1.2 volts rms, 400 cps, or 31 ± 1.2 volts (positive) shall cause the meter to register at the right edge of the red area. The input resistance at pin 21 shall be 75 ohms ± 4% at a temperature of 77° F. The Test Set shall provide at pin 4 a voltage signal at a frequency of 6150 cps ± 200 cps.

3.4.1.5.22 Switch position 22, Auxiliary Firing (AIS) - The Test Set shall measure the voltage at pin 21 on the test set, with pin 31 as ground reference. 17.5 volts rms, 400 cps or 17.5 volts dc (positive) shall cause the meter to register at the left edge of the red area. 31 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the red area. The input impedance at pin 21 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F. The Test Set will provide at pin 14, 26 VDC nominal when at this position.

3.4.1.5.23 Switch Position 23, Power Interlock (AIS) - The Test Set shall check between pins 27 and 31 (ground) for proper operation of the Power Interlock relay (in aircraft or Test Set Simulator). A closed relay shall cause the meter to register above the black area and the pilot light shall glow. An open relay shall cause the meter to register at the zero position and the pilot light shall not glow.

3.4.1.5.24 Switch Position 24, Gas Grain Stray Voltage (AIS) - The Test Set shall measure the stray voltage at pin 26 with pin 31 as ground reference. 0.1 volt dc (positive) or 0.1 volt rms, 400 cps shall cause the meter to register above the black area. 29 volts dc (positive) or 29 volts rms, 400 cps, applied for not more than 15 seconds shall not cause any damage to the equipment. The input resistance shall exceed 100 ohms. The noise pick-up in the Test Set and Cable Assy shall not cause the meter to register greater than 10% of the black area.

3.4.1.5.25 Switch Position 25, Launch Latch Stray Voltage (IS) - The Test Set shall measure the stray voltage at pin 32 on the test set, with pin 31 as ground reference. 0.1 volt dc (positive) or 0.1 volt rms, 400 cps, shall cause the meter to register above the black area. 29 volts dc (positive) or 29 volts rms, 400 cps, applied for not more than 15 seconds, shall not cause any damage to the equipment. The input resistance shall exceed 100 ohms. The noise pick-up in the Test Set and Cable Assy shall not cause the meter to register greater than 10% of the black area.

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3.4.1.5.26 Switch Position 26, Fuze Stray Voltage (A) - The Test Set shall measure the stray voltage at the forward (fuze) contact of the launcher, with pin 31 as ground reference. 0.1 volt dc (positive) or 0.1 volt rms, 400 cps, shall cause the meter to register agove the black area. 29 volts dc (positive) oror 29 volt rms, 400 cps, applied for not more than 15 seconds shall not cause any damage to the equipment. The input resistance shall exceed 100 ohms. The noise pick-up in the Test Set and Cable Assy shall not cause the meter to register greater than 10% of the black area.

3.4.1.5.27 Switch Position 27, Power Change-Over and Fuze Stray Voltage (IS) - The Test Set shall measure the stray voltage at pin 17 with pin 31 as a ground reference. 0.1 volt dc (positive) of 0.1 volt rms, 400 cps, or 29 volts rms, 400 cps, applied for not more than 15 seconds, shall not cause any damage to the equipment. The input resistance shall exceed 100 ohms. The noise pick-up in the Test Set and Cable Assy, shall not cause the meter to register greater than 10% of the black area.

3.4.1.5.28 Switch Position 28, Propulsion Stray Voltage (AIS) - The Test Set shall measure the stray voltage at the rear (propulsion) contact of the launcher, with pin 31 as a ground reference. 0.1 volt dc (positive) or 0.1 volt rms, 400 cps, shall cause the meter to register above the black area. 29 volts dc (positive) or 29 volts rms, 400 cps, applied for not more than 15 seconds, shall not cause any damage to the equipment. The input resistance shall exceed 100 ohms. The noise pick-up in the Test Set and Cable Assy shall not cause the meter to register greater than 10% of the black area.

3.4.1.5.29 Switch Position 29, Stray Voltage Circuit Check (AIS) - The Test Set shall introduce a stray voltage signal of 0.35 volts dc (positive) to the Test Set for stray voltage measurement by using pins 35 and 21. The voltage signal shall cause the meter to register in the yellow area.

3.4.1.6 Aircraft Circuit Tests: The Test Set, Guided Missile Launcher TS-1436/ASM-20 shall perform the aircraft circuit tests when a missile launcher is not attached to the aircraft. The Test Set shall measure and register the individual voltage quantity with respect to pin 31 (ground) and the specified pin of the connector J101 of the Test Set. Cable Assembly, CX6321/ASM-20 shall be the link between the Test Set and the aircraft circuitry when performing the aircraft circuit tests. Connector P101 of Cable Assembly CX-6321/ASM-20 shall be connected to J101 of the Test Set. P102 shall be connected to J102 of Adapter, Connector U214/U or U-215/U. Adapter, Connector U214/U or U-215/U shall be connected to the aircraft connector.

(1) Adapter, Connector U-214/U shall be used when testing aircraft circuits applicable to 1 and 1A missiles.

(2) Adapter, Connector U-215/U shall be used when testing aircraft circuits applicable to 1 and 1A, Mark 29 or Mark 30 missiles when using a 1C Launcher.

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(3) The "pins" referenced in the following switch positions are pins of connector J101 of the Test Set TS-1436/ASM-20.

3.4.1.6.1 Switch Position 30, 26 volt DC, or 26 volt 400 cps (Aircraft Test)(AIS) - The Test Set shall measure the voltage between pins 16 and 31. 22.5 volts rms, 400 cps or 22.5 voltd dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts rms, 400 cps or 31. 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 16 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.6.2 Switch Position 31, N2 Solenoid (Aircraft Test)(I) - The Test Set shall measure the voltage between pin 9 and pin 31. 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts dc shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 9 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.6.3 Switch Position 32, Master Arm (Aircraft Test)(AIS) - The Test Set shall measure the voltage between pin 32 and pin 31. 22.5 volts rms, 400 cps or 22.5 voltd dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 32 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.6.4 Switch Position 33, Firing (Aircraft Test)(AIS) - The Test Set shall measure a voltage between pin 26 and p9n 31. 22.5 volts rms, 400 cps or 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 21 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 26 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.6.5 Switch Position 34, Phase 1, 115 volts, rms 400 cps and 400 cps Audio Sinnal (Aircraft Test)(AIS) - The test Set shall measure the voltage between pin 3- and pin 31 and check the operation of the pilot light. 101 ± 2 volts, rms, 400 cps shall cause the meter to read at the left edge of the yellow area. 123 ± 2.5 volts, rms, 400 cps shall cause the meter to read at the right edge of theyellow area. Voltage within this range shall cause the pilot light to glow. The input resistance shall be freater than 5K ohms. Approximately 1 volt, rms, 400 cps shall be present at pin 18 when a y00 ohm load is connected between pin 18 and pin 31 on the Test Set. The output impedance at pin 18 shall be 150 ohms nominal.

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3.4.1.6.6 Switch Position 35, Phase 2, 115 volts, rms, 400 cps and 400 cps Audio Signal (Aircraft Test)(AIS) - The Test Set shall measure the voltage between pin 36 and pin 31. 101 ± 2 volts, rms, 400 cps shall cause the meter to read at the left edge of the yellow area. 123 ± 2.5 volts, rms, 400 cps shall cause the meter to read at the right edge of the yellow area. The input resistance shall be breater than 6K ohms. Approximately 1V, rms, 400 cps shall be present at pin 18 when a 600 ohm load is connected between pin 18 and pin 31 on the Test Set. The output impedance at pin 18 shall be 150 ohms nominal.

3.4.1.6.7 Switch Position 36, Phase 3, 115 volts, rms, 400 cps and 400 cps Audio Signal (Aircraft Test)(AIS) - The Test Set shall measure the voltage between pin 37 and pin 31. 101 ± 2 volts, rms, 400 cps shall cause the meter to read at the left edge of the yellow area. 123 ± 2.5 volts, rms, 400 cps shall cause the meter to read at the right edge of the yellow area. The input resistance shall be greater than 6K ohms. Approximately 1 volt, rms, 400 cps shall be present at pin 18 when a 600 ohm load is connected between pin 18 and pin 31 on the Test Set. The output impedance at pin 18 shall be 150 ohms nominal.

3.4.1.6.8 Switch Position 37, Ground Check (Aircraft Test)(AI) - The Test Set will check for a ground on pin 24 of the aircraft power supply applying a 100 microamp reference current to the test set meter on position 42 through pin 13. If pin 24 in the aircraft power supply is grounded, the Test Set will read in the yellow area. If pin 24, the Aircraft single Phase 115V, 400 cps, is not grounded, the meter will read in the black area.

3.4.1.6.9 Switch Position 38, Jettison (Aircraft Test)(AIS) - The Test Set shall measure the voltage between pin 17 and pin 31. 22.5 volts rms, 400 cps or 22.5 volts dc (positive) shall cause the meter to register at the left edge of the yellow area. 31 ± 1.2 volts rms, 400 cps or 31 ± 1.2 volts dc (positive) shall cause the meter to register at the right edge of the yellow area. The input resistance at pin 17 shall be 75 ohms ± 4% at a temperature of 77° F ± 10° F.

3.4.1.6.10 Switch Position 39, Tunging Motor and AFC Meter (Aircraft Test)(S) - The Test Set shall measure a voltage of 10.90 volts rms, 400 cps at pin 11 with pin 31 as ground refernnce. 8.75 volts rms, 400 cps shall cause the meter to register at the left edge of the yellow area. 11.93 volts ± 0.5 volts rms, 400 cps shall cause the meter to register at the right edge of theyellow area. The input resistance at pin 11 shall be 260 ohms at a temperature of 77° f ± 10° F. The input at position 39 is caused by an output which is present on pin 23 of the Test Set, when the selector switch is on position 39. The output signal shall be 0.87 volt peak-to-peak square wave, 400 cps, superimposed on 4.90 volts dc nominal. The load appearing in the launcher will be 3000 ohms ± 10% to the ac signal and 5000 ohms ± 5% to the dc signal.

3.4.6.11 Switch Position 40, Meter Test (Aircraft Test)(AIS) - The Test Set shall introduce a stray voltage signal of 0.35 solts dc (positive) to the Test Set for stray voltage measurement by using pins 35 and 21. The voltage signal shall cause the meter to register in the yellow area.

The meter shall operate satisfactorily under

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3.4.1.6.12 Switch Position 41, 3 Phase, 115V, 400 cps, Phase Sequence Check (Aircraft Test)(S) - The Test Set shall test for positive phase sequence using lines 30, 36 and 37. A three phase input of 101 VAC ± 2 VAC rms, line-to-ground, positive phase sequence (phase 1 at pin 30, phase 2 at pin 36 and phase 3 at pin 37) shall cause the meter to read at the left edge of the yellow area. A three phase input of 123 ± 2.5 VAC rms, line-to-ground, positive phase sequence shall cause the meter to read in the yellow area. A reverse or negative phase sequence shall cause the meter to read below the yellow area.

3.4.1.6.13 Switch Position 42, Power Interlock (Aircraft Test) (AIS) - The test Set shall check between pins 27 and 31 (ground) for proper operation of the Power Interlock relay. A closed relay shall cause the meter to read above the black area and the pilot light shall glow. An open relay shall cause the meter to read at the zero position and the pilot light shall not glow.

3.4.1.7 Missile ID Power Supply Tests - The Test Set Guided Missile Launcher TS-1436/ASM-20 shall perform tests utilizing the Millile ID relays. The test set-up is similar to that specified in 3.4.1.6.

3.4.1.7.1 IR Position (Mark 29) - When the Missile ID Switch is in the IR position, +26 VDC ± 4 volts shall be provided at pin 1 of the Test Set with a load of 150 milliamperes. The input at pin 33 of the Test Set to supply this dc voltage will be 26 VDC.

3.4.1.7.2 1 and 1A Position (Mark 2) - When the Missile ID Switch is in the 1 and 1A position, no voltage shall be providee at pin 1.

3.4.1.7.3 SR Position (Mark 30) - When the Missile ID Switch is in the SR position, -26 VDC ± 4 volts shall be provided at pin 1 of the Test Set with a load of 150 milliamperes. The input at pin 30 of the Test Set to supply this dc voltage will be 115 VAC rms, 400 cps. The -26 VDC power supply return is on pin 20.

3.4.1.7.4 Missile ID Power Supply Protection - The power supplies will be protected with a 0.5 ampere automatic circuit breaker. This circuit breaker shall be in series with pin 1 of the Test Set and the Missile ID switch.

3.4.2 Cable, Special Purpose, Electrical CX-6321/ASM-20 - The Special Purpose Cable shall conform to the applicable requirements of Specification MIL-T-21200 and the following:

3.4.2.1 Function - The cable CX-6321/ASM-20 shall perform the electrical connections between the aircraft missile launcher and the Guided Missile Launcher Test Set TS-1436/ASM-20.

3.4.2.2 Form Factor - The cable CX-6321/ASM-20 shall contain 37 wires and be terminated by a Connector Deutsch Type DM-9718-375-1D (or approved equivalent) or one end and a connector Deutsch Type DM-9718-37P-1D (or approved equivalent) on the other end. A special connector is provided on the cable for checking the launcher firing circuit. The cable shall not exceed 7/8 inches in diameter and 15½ feet in length.

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3.4.2.3 Weight - The weight of the cable assembly shall not exceed 9.0 pounds.

3.4.3 Adapter, Connector - The Adapter, Connectors shall meet the following requirements, and conform to the applicable requirement of MIL-T-21200.

3.4.3.1 Form Factor and Weight - The Adapter, Connectors shall conform to the following:

Designation	BUWEPS Dwg. No.	Size (inches) W	H	L	Weight
U-213/U	60A54C81	1½	1½	4¼	5/8 pound
U-214/U	60A54C87	1½	1½	4 3/8	5/8 pound
U-215/U	60A54C92	1½	1½	4¼	5/8 pound
U-216/U	60A54C98	1½	1½	3¼	1/2 pound

3.4.3.2 Function - The Adapter, Connectors shall perform the following functions:

3.4.3.2.1 Adapter U-213/U - This adapter is used to test those launcher circuits applicable to 1 and 1A missiles.

3.4.3.2.2 Adapter U-214/U - This adapter is used to test those aircraft circuits applicable to 1 and 1A missiles.

3.4.3.2.3 Adapter U-215/U - This adapter is used to test those aircraft circuits applicable to 1 and 1A, MK29(IR) or MK30(SR) missiles when using a 1C Launcher.

3.4.3.2.4 Adapter U-216/U - This adapter is used to test those laucher circuits applicable to a MK29(IR) or MK30(SR) missile.

3.4.4 Case, Test Set, CY-2943/ASM-20 - The Case, Test Set, CY-2943/ASM-20 shall be a transit case made in accordance with BUWEPS Drawing No. 60A54D6 and shall conform to the applicable requirements of Specification MIL-T-21200.

3.4.4.1 Function - The Case, Test Set, CY-2943/ASM-20 shall provide stowage for the TS-1436/ASM-20, Cable CX-6321/ASM-20, four Adapters (U-213 through U-216/U), Pressure Gauge MX-3298/ASM-20, two special wrenches and operating manual.

3.4.4.2 Form Factor - The Case, Test Set CY-2943/ASM-20 shall have the maximum dimensions of 11 1/8 inches wide, by 8 5/8 inches high, by 17 inches long.

3.4.4.3 Weight - The Case, Test Set CY-2943/ASM-20 shall not exceed 13 pounds.

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3.4.5 Gauge, Pressure, Dial Indicating - MX3298/ASM-20) - The Gauge, Pressure, Dial Indicating, MX-3298/ASM-20 shall be made in accordance with BUWEPS Drawing No. 60A54D116 and conform to the applicable requirements of Specification MIL-T-21200.

3.4.5.1 Function - The Gauge is used to perform a static pressure check of the nitrogen system in the missile launcher for the MK29(IR) missile. The Pressure Tester has colored areas indicating pressure ranges. The electrical interlock in the Pressure Tester connector completes the nitrogen solenoid circuit.

3.4.5.2 Form Factor and Weight - The gauge shall have maximum dimensions of 1½ inches wide by 1½ inches high and 4½ inches long with a maximum weight of ½ pound.

3.4.6 Wrenches, HEX - The two special hex wrenches are used to raise the detent in the missile launcher rail to allow insertion of the lug handle of the cable assembly into the launcher rail, and for opening some launcher nose covers. The 5/8 inch wrench is also used as a safety pin during testing of the 1C launcher.

3.4.6.1 Form Factor and Weight - The Hex wrenches shall have the following sizes and weights:

(1) Size 3/8 inch, drawing No. 58A5A294-12, 4 3/4 inches wide by 1 7/8 inches long, weight 1/4 pound.

(2) Size 5/16 inch, Drawing No. 60A54C5, 5 5/8 inches wide by 5 1/8 inches long, weight 5/16 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as othersise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.4.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

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(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the arproved test procedure of 4.5. The government inspector and the procuring activity shall be advised wher tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Service Approval Tests - At the completion of contractor demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental testing in accordance with Specification MIL-T-5422 , including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall by by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute

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cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent by the contractor at his expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequancy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be birated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractural action)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, and special tests.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094, Procedure I. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Test Level - The Test level shall be Test Level II (from Table I of Specification MIL-R-23094). During each cycle the equipment shall be operated at the following time(s):

4.4.3.2 ACCEPT-Reject Criteria - Figure 16 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.3.3 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated 1 hours during each cycle.

4.4.3.4 Performance Characteristics to be Measured - The performance characteristics to be measured during the reliability tests shall be submitted by the contractor to the procuring activity for review and approval prior to starting reliability tests as part of the reliability test procedure.

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4.4.3.5 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

1. Whenever performance characteristics fall below the Acceptance requirement at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests no preventive measures may be performed upon the equipment:

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity. The test procedure for the reliability tests shall be prepared by the contractor and approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Guided Missile Launcher Test Set, AN/ASM-20, is intended for safe operating tests on those launchers which launch the Sidewinder Missiles, Mark 2, Mark 29 or Mark 30.

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6.2 Ordering Data - Purchasers should exercise any desired options herein and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete information.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (6-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R308

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in a window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

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